Then $0 \le G(x) \le 1$ $\forall x$ and G is non decreasing.

Proof for non-decreasing part:

If $x < y$, then $F_n(x) \le F_n(y)$

$\Rightarrow \lim_{n\to\infty} F_n(x) \le \lim_{n\to\infty} F_n(y)$

Thm: Helly's selection theorem

Suppose $\{F_n\}_n$ is a seq of DFs.
Then $\exists$ a subsequence $\{F_{n_k}\}_k$ and a non-decreasing, right continuous function $F: \mathbb{R} \to [0,1]$ such that $\lim_{k\to\infty} F_{n_k}(x) = F(x)$ $\forall x$, where F is continuous. (i.e. the points where limit function has jumps) $\longrightarrow$ (continuity points)

$\lim_{n\to\infty} F_{X_n}(x) = \begin{cases} 0 & x<0 \\ 1 & x>0 \end{cases} \longrightarrow \lim_{n\to\infty} F_{X_n}(x) = \begin{cases} 0, & x<0 \\ 1, & x\ge 0 \end{cases}$

but Ex2 & Ex3 $\lim$ function can never be CDF.

Weak convergence of RVs (convergence in law) or (convergence in Distribution)

Suppose $X_n : (\Omega_n, \mathcal{F}_n, \mathbb{P}_n) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_n X_n^{-1})$ r.v.
RVs and Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}X^{-1})$ be another RV. Let F_{X_n} denote the DF of X_n, F_X denote the DF of X.

We say that $\{X_n\}_n$ converges in law/ distribution to X if:

$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$ at all continuity points x of F_X.

(#)

Ex1: $X_n \sim d_{1/n}$, $X \sim d_0$.

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1, & x > 0 \\ 1, & x = 0 \end{cases}$$

$x < 0 \longleftrightarrow \lim_{n\to\infty} F_{X_n}(x) = 0$

$x > 0 \longleftrightarrow \lim_{n\to\infty} F_{X_n}(x) = 1$

$x = 0$: ignore jumps.

Notation:

We denote the convergence in (*) by $F_{X_n} \Rightarrow F_X$. alternatively $X_n \Rightarrow X$

$$X_n : (\Omega_1, \mathcal{F}_1, \mathbb{P}_n) \longrightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mathbb{P}_n \circ X_n^{-1})$$

$$X : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mathbb{P}\circ X^{-1})$$

Weak convergence of probability measures on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$

Let $\{\mu_n\}_n$ be a sequence of probability measures on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ with DF $F_n : \mathbb{R} \to [0,1]$

$$F_n(x) = \mu_n((-\infty, x])$$

We say $\{\mu_n\}_n$ converges weakly to a probability measure μ, denoted by: $\mu_n \Rightarrow \mu$ if:

$\lim_{n\to\infty} F_n(x) = F(x)$ at all continuity points x of DF F of μ.

Equivalently: $\mathbb{P}_n \circ X_n^{-1} \Rightarrow \mathbb{P}\circ X^{-1}$

$F_{X_n} \Rightarrow F_X$

$X_n \Rightarrow X$

$\mathbb{P}_n \circ X_n^{-1} \Rightarrow \mathbb{P}\circ X^{-1}$

equivalent.

domain is not important, look at range side and look at convergence

Thm:

This result is a combination of :

(i) Le'vy's continuity thm.

(ii) Portmanteau Theorem / Lemma.

(iii) Helly – Bray Thm.

The following are equivalent.

I. $X_n \Rightarrow X$ (i.e. $F_{X_n} \Rightarrow F_X$)

II. For all $f : \mathbb{R} \to \mathbb{R}$ bounded & continuous we have $\lim_{n\to\infty} \mathbb{E} f(X_n) = \mathbb{E}(f(X))$

III. $\forall x \in \mathbb{R}$ $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$.

$$\mathbb{E} f(X)$$

$$= \int_{\Omega} f(X(\omega))\, d\mathbb{P}(\omega)$$

$$= \int f(x)\, \underbrace{d\mathbb{P}\circ X^{-1}(x)}_{\uparrow \text{law of } X}.$$